Healthcare Data Analysis: Predicting Diabetes Risk

1. Project Group

- **Project Members:**

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2. Project Topic

Application Subject Area

- **Domain:** Healthcare
- **Objective:** Develop a predictive model to identify individuals at risk of diabetes using machine learning techniques in R.

Dataset and Sources

- **Dataset:** PIMA Indians Diabetes Dataset
- **Source:** UCI Machine Learning Repository (https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database/data)
- **Description:** The dataset consists of medical diagnostic measurements and personal details of patients, including:
  - Pregnancies
  - Glucose levels
  - Blood Pressure
  - Skin Thickness
  - Insulin
  - BMI
  - Diabetes Pedigree Function
  - Age
  - Outcome (Diabetes: 1, No Diabetes: 0)

Reference Resources

- Research papers on diabetes prediction using ML techniques.
- Articles discussing feature importance in medical predictions.
- Documentation on machine learning in R (caret, randomForest, glm packages).

Supplemental Resources

- **R Libraries Used:**
    - tidyverse – Data manipulation and visualization
    - caret – Machine learning framework
    - randomForest – Random forest classifier
    - ggplot2 – Data visualization
    - e1071 – SVM for model comparison

3. Project Methodology

Data Preprocessing

- Handle missing values using mean imputation.
- Normalize numerical features.
- Split the dataset into training (80%) and testing (20%) sets.

Exploratory Data Analysis (EDA)

- Visualize data distribution using ggplot2.
- Check correlation between features using cor() function.

Model Building

- Implement Logistic Regression, Random Forest, and Support Vector Machine (SVM) models.
- Perform hyperparameter tuning using caret:: trainControl().
- Evaluate models using accuracy, precision, recall, and F1-score.

Results and Discussion

- Compare model performances.
- Identify key features contributing to diabetes risk.
- Provide insights for healthcare professionals.

Conclusion and Future Work

- Summarize findings and accuracy of models.
- Suggest improvements, such as collecting more data or using deep learning techniques.